

AI-Powered Clinical Decision Support & Digital Health Record Platform

1. Executive Summary

This startup aims to build an AI-powered healthcare platform that combines Electronic Health Records (EHR), real-time doctor-patient interaction analysis, and AI-assisted diagnosis and prescription generation. Each patient will have a unique digital profile containing medical history, lab reports, vitals, and consultation records. The system will assist doctors by generating tentative diagnoses and prescriptions, improving speed, accuracy, and efficiency in clinical decision-making.

2. Problem Statement

- Patient data is fragmented across clinics and labs
- Doctors lack complete medical history during consultations
- Diagnosis depends heavily on time and manual interpretation
- High patient load reduces consultation quality
- Limited use of AI in small and mid-sized clinics

3. Solution

A centralized AI-driven healthcare platform that creates a unique digital profile for every patient, enables real-time data updates, analyzes doctor-patient conversations, and suggests probable diagnoses, required tests, and draft prescriptions. The doctor reviews and finalizes all outputs.

4. Product Features

Patient Management System

- Unique patient ID
- Full medical history tracking
- Lab report uploads and integrations

Role-Based Access

Compounder/Nurse Panel

- Upload vitals
- Update patient data

Doctor Panel

- Access complete records
- AI-assisted diagnosis
- Edit and finalize prescriptions

AI Engine

- Speech-to-text
- Symptom extraction (NLP)
- Diagnosis recommendation
- Prescription draft generation

- Risk alerts

Lab Integration

- Automatic report uploads
- API-based lab integration

5. Target Market

Primary Customers

- Small and medium clinics
- Private practitioners
- Diagnostic centers
- Telemedicine providers

Secondary Market

- Hospitals
- Health-tech platforms
- Insurance companies

6. Revenue Model

- 1 SaaS Subscription: Monthly fees per clinic (₹999 – ₹4999)
- 2 Pay-per-consultation AI fee
- 3 Lab partnerships and commissions
- 4 Premium AI features and analytics

7. Competitive Advantage

- EHR + AI + Voice Analysis
- Designed for Indian healthcare workflows
- Supports doctors, does not replace them
- Highly scalable

8. Technology Architecture

- Frontend: React.js (Web), React Native (Mobile)
- Backend: Node.js / Django
- Database: PostgreSQL, MongoDB
- AI Stack: LLM APIs, Speech Recognition, NLP
- Cloud: AWS / Azure / GCP

9. Go-To-Market Strategy

- 1 Pilot launch with 5–10 clinics
- 2 Expansion to Tier-2 and Tier-3 cities
- 3 Scaling via hospital and insurance partnerships

10. Development Roadmap

- 1 MVP (2–3 months): Core system and data management
- 2 AI Integration (2 months): Speech and diagnosis
- 3 Advanced Features (3 months): Prescription and lab integration
- 4 Scaling: Mobile apps and analytics

11. Regulatory & Compliance

- Data privacy and encryption
- HIPAA-like compliance standards
- Audit logs
- Clear labeling of AI-generated suggestions

12. Risks & Challenges

- Incorrect AI suggestions
- Data privacy risks
- Doctor adoption resistance
- High development complexity

13. Future Expansion

- Predictive analytics
- Chronic disease monitoring
- Insurance integration
- Remote patient monitoring
- AI-powered telemedicine

14. Team Requirements

- 2–3 Software Developers
- 1 AI/ML Engineer
- 1 UI/UX Designer
- 1 Medical Advisor
- 1 Product Manager